Chapter 2: Motion in a Straight Line

1. FUNDAMENTAL KINEMATIC PARAMETERS

Instantaneous Velocity & Acceleration

MUST KNOW

$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}, \quad a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = v\frac{dv}{dx} = \frac{d^2x}{dt^2}$$

Symbols: x = position, v = instantaneous velocity, a = acceleration, t = time.

SI Units: $v \to \mathrm{m\,s^{-1}}$, $a \to \mathrm{m\,s^{-2}}$

Conditions: $a = v\frac{dv}{dx}$ is specifically used when acceleration is expressed as a function of position x rather than time t.

CAUTION: ZERO VELOCITY VS. ZERO ACCELERATION

$v = 0$ at an instant does **NOT** mean $a = 0$. Example: At the apex of vertical throw, $v = 0$ but $a = -g = -9.8\ \mathrm{m\,s^{-2}}$.

2. UNIFORMLY ACCELERATED RECTILINEAR MOTION (a = constant)

Standard Equations of Motion

MUST KNOW EXAM FAVOURITE

$$v = v_0 + at, \quad x = x_0 + v_0 t + \frac{1}{2}at^2, \quad v^2 = v_0^2 + 2a(x - x_0)$$

Symbols: v_0 = initial velocity, v = final velocity, x_0 = initial position, x = final position.

Average Velocity: $\bar{v} = \frac{v+v_0}{2}$ (Valid *only* for constant a).

Displacement in n^{th} second: $D_n = v_0 + \frac{a}{2}(2n-1)$ [Unit: m]

Calculus Derivation of Motion Equations (a = constant)

1. **Derivation of $v = v_0 + at$:**
Start with definition $a = \frac{dv}{dt} \implies dv = a\,dt$. Integrate both sides with limits $(v_0 \to v)$ and $(0 \to t)$:

$$\int_{v_0}^{v} dv = a\int_0^t dt \implies [v]_{v_0}^{v} = a[t]_0^t \implies \boxed{v = v_0 + at}$$

2. **Derivation of $x - x_0 = v_0 t + \frac{1}{2}at^2$:**
Substitute $v = \frac{dx}{dt} \implies dx = v\,dt = (v_0 + at)\,dt$. Integrate both sides with limits $(x_0 \to x)$ and $(0 \to t)$:

$$\int_{x_0}^{x} dx = \int_0^t (v_0 + at)\,dt \implies x - x_0 = v_0[t]_0^t + a\left[\frac{t^2}{2}\right]_0^t \implies \boxed{x = x_0 + v_0 t + \frac{1}{2}at^2}$$

3. **Derivation of $v^2 = v_0^2 + 2a(x - x_0)$:**
Use chain rule $a = v\frac{dv}{dx} \implies v\,dv = a\,dx$. Integrate both sides with limits $(v_0 \to v)$ and $(x_0 \to x)$:

$$\int_{v_0}^{v} v\,dv = a\int_{x_0}^{x} dx \implies \left[\frac{v^2}{2}\right]_{v_0}^{v} = a[x]_{x_0}^{x} \implies \frac{v^2 - v_0^2}{2} = a(x - x_0) \implies \boxed{v^2 = v_0^2 + 2a(x - x_0)}$$

3. VERTICAL MOTION UNDER GRAVITY (FREE FALL)

Free Fall Equations (Sign Convention: Upward = $+y$) MUST KNOW

$$a = -g \implies v = v_0 - gt, \quad y = y_0 + v_0 t - \frac{1}{2}gt^2, \quad v^2 = v_0^2 - 2g(y - y_0)$$

Max Height ($v = 0$): $H_{\max} = \frac{v_0^2}{2g}$

Time to Top: $t_{\text{up}} = \frac{v_0}{g}$

Total Time of Flight: $T = \frac{2v_0}{g}$

Impact Velocity: $v_{\text{impact}} = -v_0$

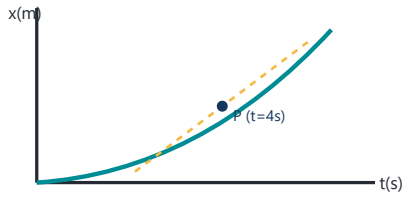
SHORTCUT: GALILEO'S LAW OF ODD NUMBERS

For a body dropped from rest ($v_0 = 0$) under gravity, distances traversed in equal successive time intervals τ follow the odd number ratio:

$$y_1 : y_2 : y_3 : y_4 : \cdots = 1 : 3 : 5 : 7 : \ldots \quad \text{where } y_1 = \frac{1}{2}g\tau^2$$

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4. GRAPHICAL ANALYSIS & PHYSICAL MEANINGS



Position-Time Graph (

$$x - t$$

)

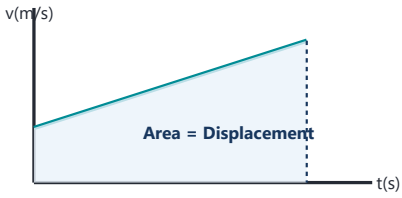
- **Slope of tangent:** Instantaneous velocity (

$$v = \frac{dx}{dt}$$

)
- **Curvature upward:** Positive acceleration (

$$a > 0$$

)



Velocity-Time Graph (

$$v - t$$

)

- **Slope of line:** Acceleration (

$$a = \frac{dv}{dt}$$

)
- **Area under curve:** Net Displacement (

$$\Delta x = \int v\,dt$$

)

5. NUMERICAL SOLVING TEMPLATES

1. Stopping Distance & Reaction Time Problems

Given: $v_0, a = -d$ → Formula: $v^2 = v_0^2 + 2ax$ $(v = 0)$ → Stopping Distance: $d_s = \frac{v_0^2}{2a}$

- **Reaction Time (t_r):** Distance traveled during thinking phase before applying brakes = $d_{\text{think}} = v_0 \cdot t_r$
- **Total Stopping Distance:** $d_{\text{total}} = (v_0 \cdot t_r) + \frac{v_0^2}{2a}$

2. Ball Thrown Up from Building Height h (Single-Step Method)

Given: $y_0 = +h, y = 0, v_0 = +u, a = -g$ → Substitute into: $y = y_0 + v_0 t + \frac{1}{2}at^2$ → Solve: $0 = h + ut - \frac{1}{2}gt^2$

Worked Numerical Example (NCERT Ex 2.3):
Given: $y_0 = 25\ \text{m}, v_0 = +20\ \text{m s}^{-1}, a = -10\ \text{m s}^{-2}, y = 0$ (ground).

$$0 = 25 + 20t - 5t^2 \implies 5t^2 - 20t - 25 = 0 \implies t^2 - 4t - 5 = 0 \implies (t-5)(t+1) = 0$$

$$\text{Discard negative time } t = -1\text{s} \implies \boxed{t = 5\ \text{s}}$$

6. RELATIVE VELOCITY IN ONE DIMENSION

Relative Velocity Equations

MUST KNOW EXAM FAVOURITE

$$v_{AB} = v_A - v_B, \quad v_{BA} = v_B - v_A = -v_{AB}$$

Same Direction: $v_{\text{rel}} = |v_A - v_B|$ **Opposite Direction:** $v_{\text{rel}} = v_A + v_B$

Relative Acceleration: $a_{AB} = a_A - a_B \implies x_{\text{rel}} = (v_{AB})t + \frac{1}{2}(a_{AB})t^2$

SHORTCUT: POLICE VAN & THIEF BULLET CATCH PROBLEM (NCERT EX 2.14)

Formula: v_{muzzle} is bullet speed relative to police van (v_P).

- Absolute speed of bullet: $v_b = v_{\text{muzzle}} + v_P$
- Speed with which bullet hits thief's car (v_T): $v_{\text{rel}} = v_b - v_T = (v_{\text{muzzle}} + v_P) - v_T$

7. SI UNITS, DIMENSIONS & CONSTANTS REFERENCE

Physical Quantity	Symbol	Dimension	SI Unit	Key Proportionality / Constant
Displacement / Position	x, y, s	$[\text{L}]$	m	$x \propto t^2$ (for constant a, $v_0 = 0$)
Velocity	v, u	$[\text{L T}^{-1}]$	m s^{-1}	$1\ \text{km h}^{-1} = \frac{5}{18}\ \text{m s}^{-1} \approx 0.278\ \text{m s}^{-1}$
Acceleration	a, g	$[\text{L T}^{-2}]$	m s^{-2}	$g \approx 9.8\ \text{m s}^{-2} \approx 10\ \text{m s}^{-2}$
Stopping Distance	d_s	$[\text{L}]$	m	$d_s \propto v_0^2$ (Doubling v_0 quadruples d_s)